

SHETH L.U.J & SIR M.V COLLEGE
Data Visualization with Power BI and Tableau

Practical No: 8

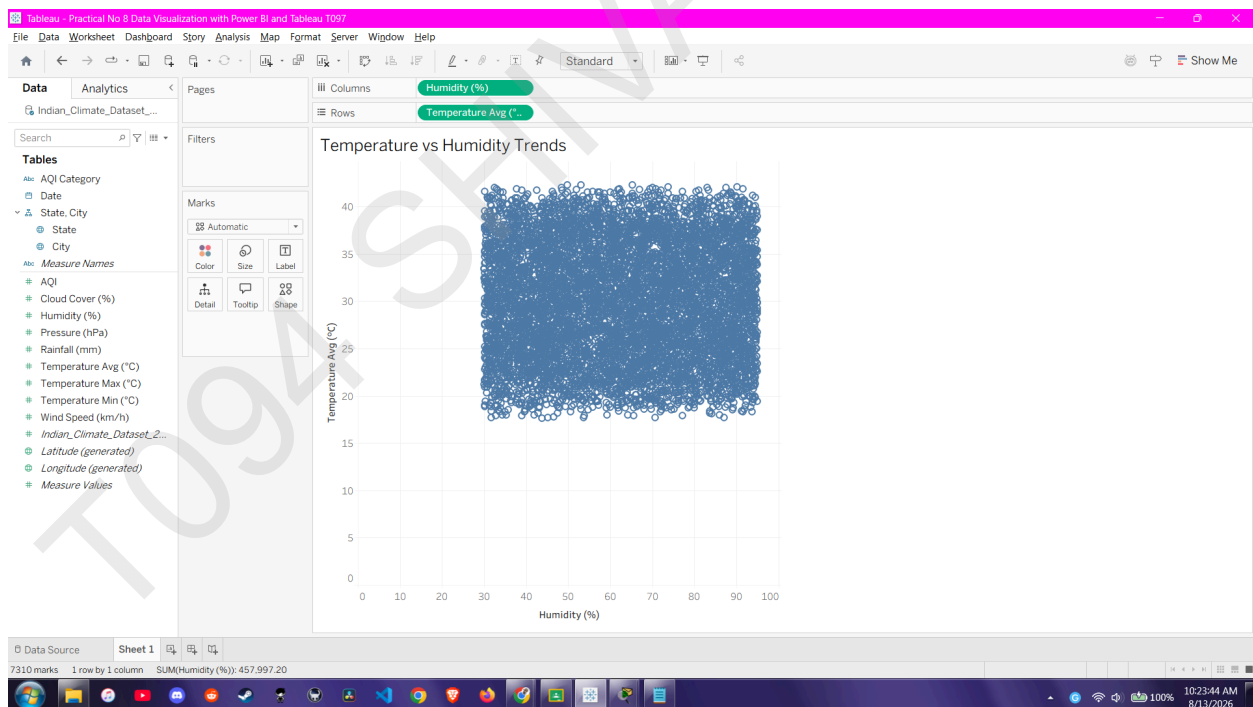
Aim: Performing Advanced Analytics using Trends, Clustering, and Forecasting

- a) Add and interpret trend lines (linear, exponential)
- b) Customize and analyze trend models
- c) Perform clustering on datasets
- d) Analyze distributions using built-in tools
- e) Apply forecasting techniques
- f) Compare different statistical models
- g) Export and interpret statistical results

a) Add and interpret trend lines (linear, exponential)

Task 1: Add a Linear Trend Line to a Scatter Plot

- Create **Sheet 1**. Drag **Humidity (%)** to the **Columns** shelf and **Temperature Avg (°C)** to the **Rows** shelf. Both should be aggregated as dimensions or disaggregated (Analysis $\rightarrow$ uncheck Aggregate Measures) to create a scatter plot.
- Click on the **Analytics** pane (next to the Data pane on the left).
- Under the *Model* section, drag **Trend Line** onto the canvas.
- Drop it onto the **Linear** model option that appears in the drop zone.



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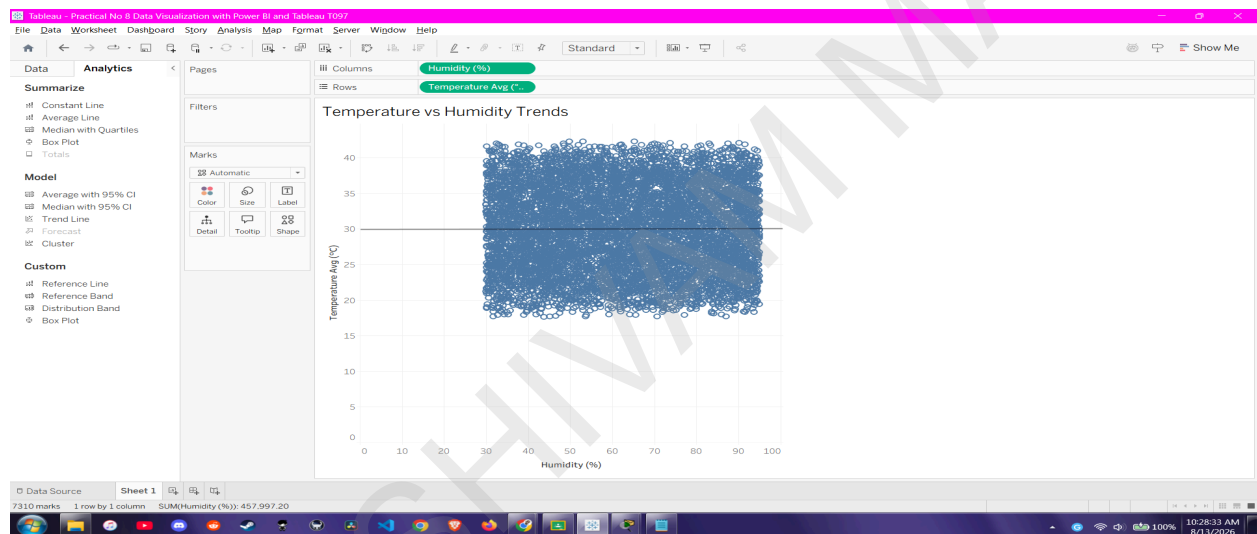
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b) Customize and analyze trend models

Task 1: Change to Polynomial and View Model Details

- Create **Sheet 1 (2)**. Repeat the setup from Task 1 (Scatter plot of Humidity vs Avg Temperature).
- Drag **City** to the **Color** tile to generate multiple trend lines.
- Right-click on any trend line and select **Edit Trend Lines**. Change the Model type from Linear to **Polynomial** with a degree of 2 or 3.
- Right-click the trend line again and select **Describe Trend Line...** to view the equation and p-value.



Task 2: View Full Trend Model Description

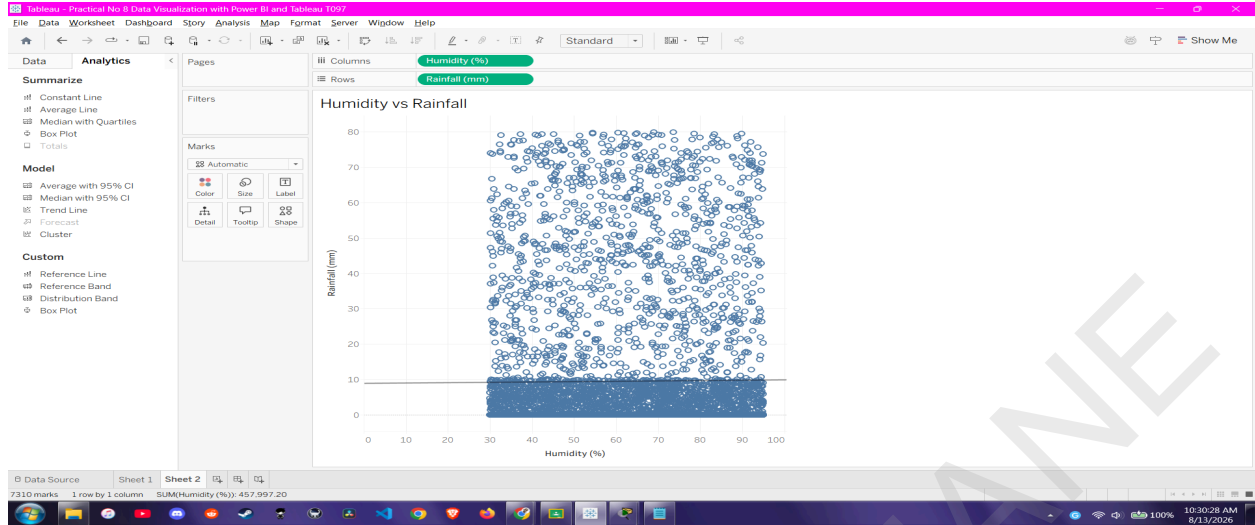
- Right-click the trend line and select **Describe Trend Model...**
- This window provides detailed statistics including R-Squared, Standard Error, and Analysis of Variance (ANOVA).

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c) Perform clustering on datasets

Task 1: Cluster Analysis on Scatter Plot

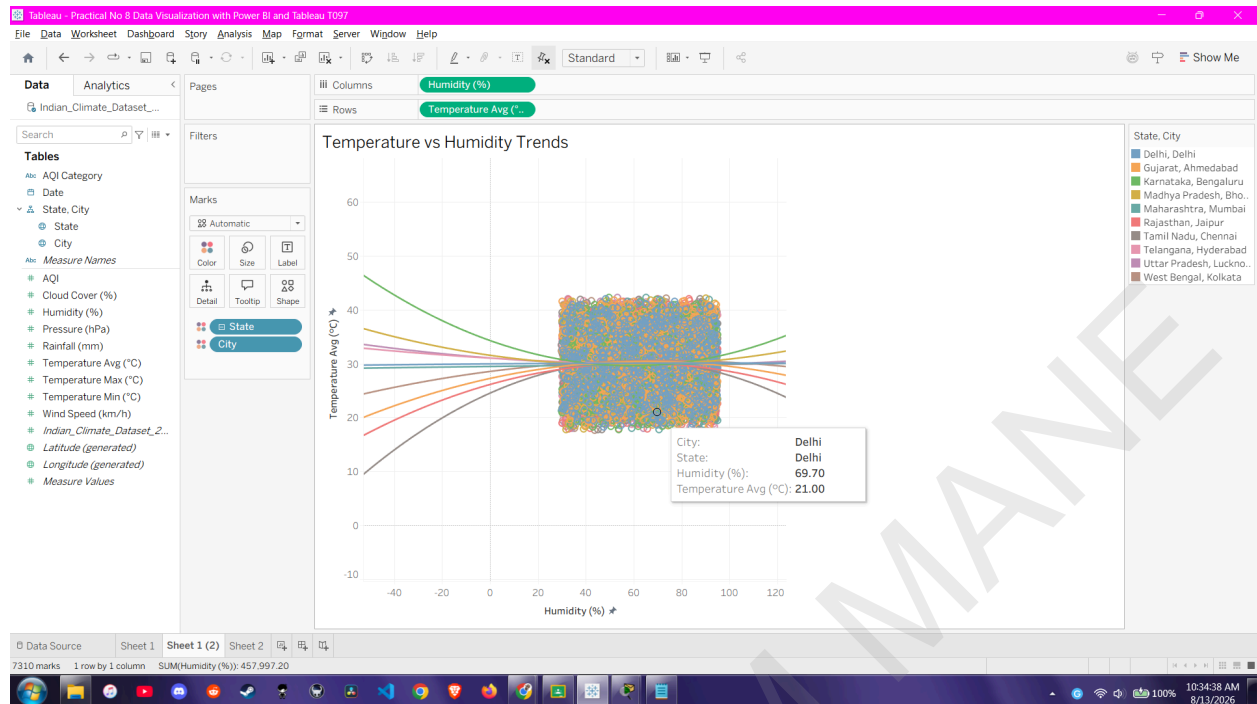
- Create **Sheet 4**. Drag **AQI** to the **Columns** shelf and **Temperature Avg (°C)** to the **Rows** shelf. Uncheck Aggregate Measures.
- Switch to the **Analytics** pane.
- Under the *Model* section, drag **Cluster** onto the scatter plot.
- In the Clusters dialog box, you can let Tableau automatically determine the number of clusters or specify a number (e.g., 10).

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Task 2: Describe Clusters

- Right-click the **Clusters** pill that was automatically added to the Color tile and select **Describe Clusters...**
- This provides a summary of the inputs, scaling, number of points per cluster, and sum of squares statistics.

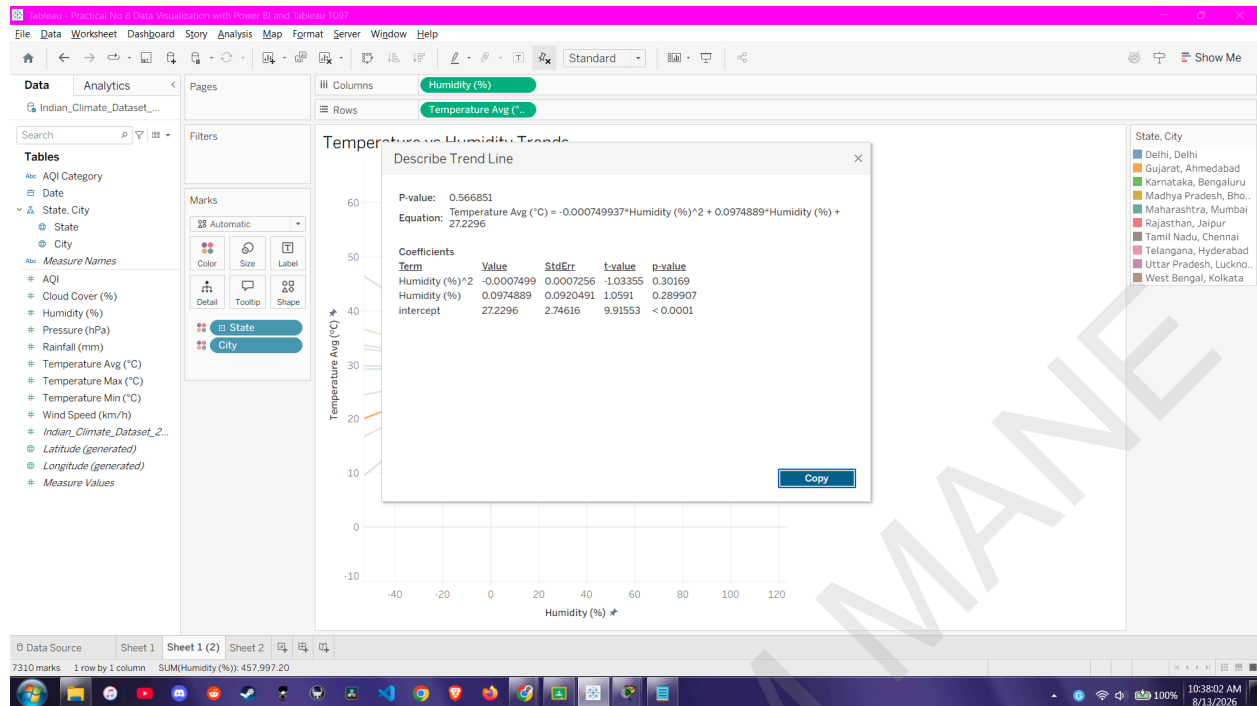
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d) Analyze distributions using built-in tools

Task 1: Add a Distribution Band

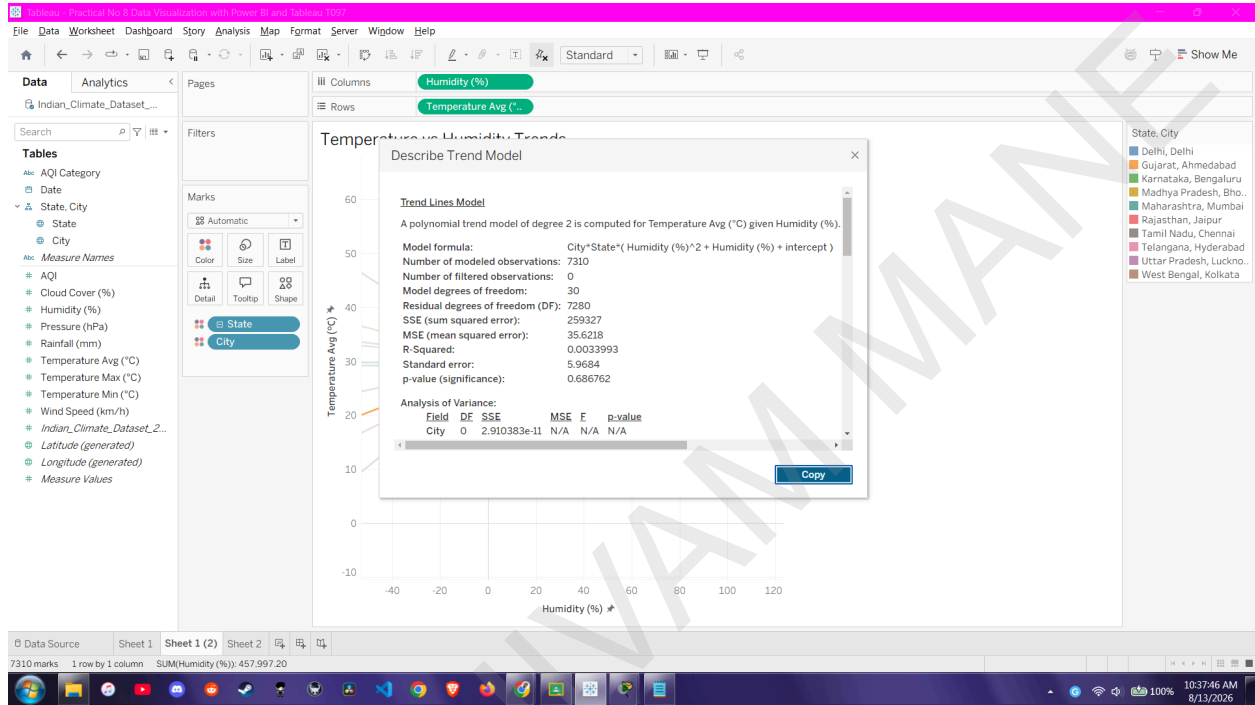
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- Create Sheet 5. Drag **City** to the Columns shelf and **Temperature Max (°C)** to the Rows shelf.
- Switch to the Analytics pane.
- Under the *Custom* section, drag **Distribution Band** onto the canvas and drop it on the Cell option.
- Configure the band (e.g., 60%, 80% of Average).



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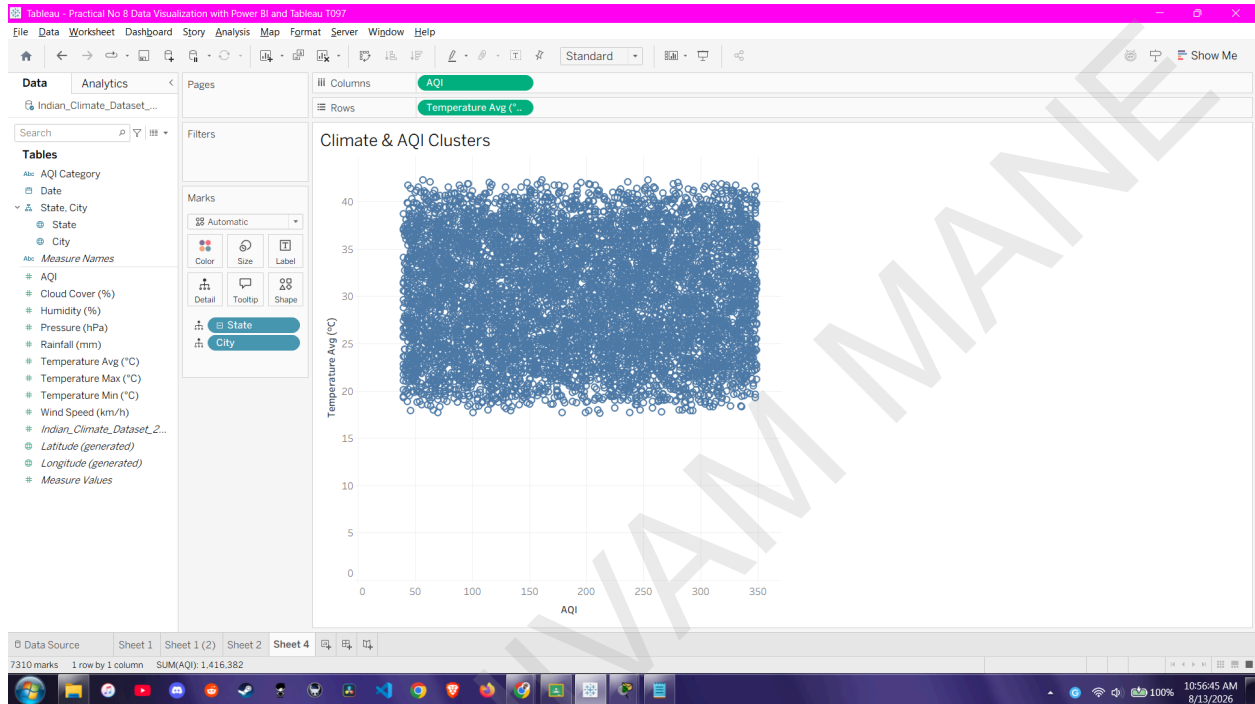
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Task 2: Create a Histogram

- Create **Sheet 6**. In the Data pane, highlight **AQI**.
- Click **Show Me** and select the **Histogram** icon.
- Tableau will automatically create **AQI (bin)** on Columns and **CNT(AQI)** on Rows, displaying the frequency distribution.



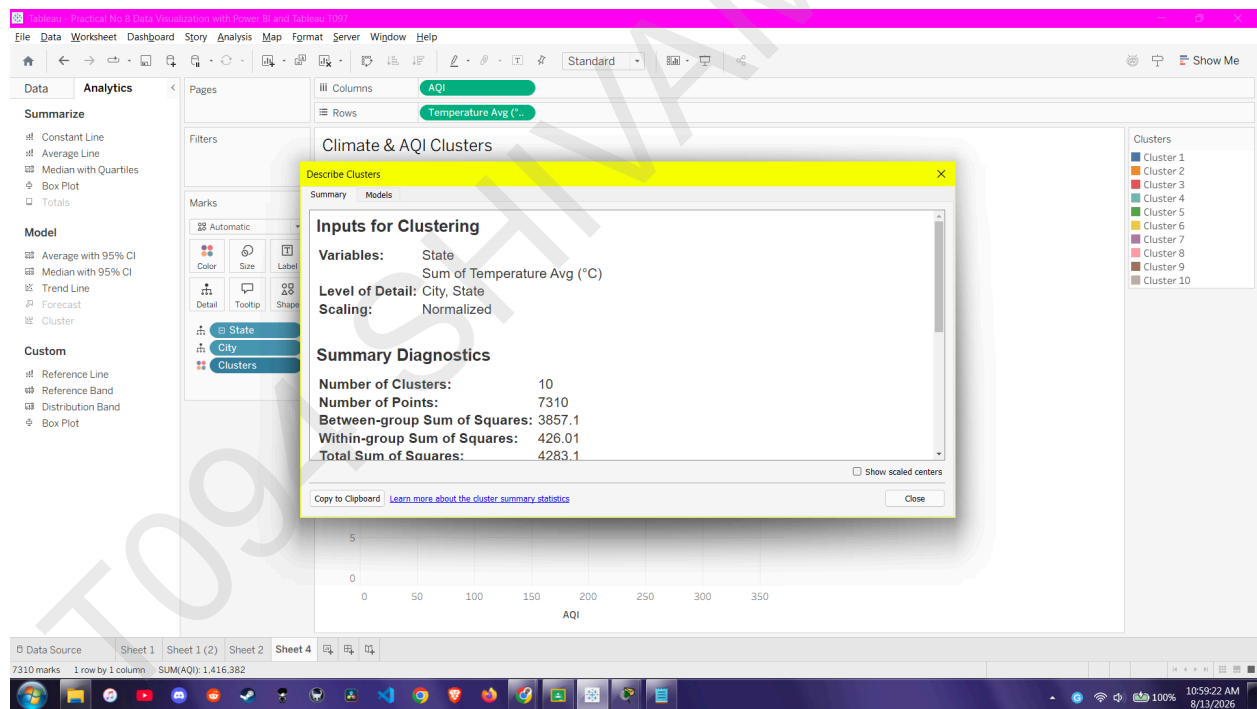
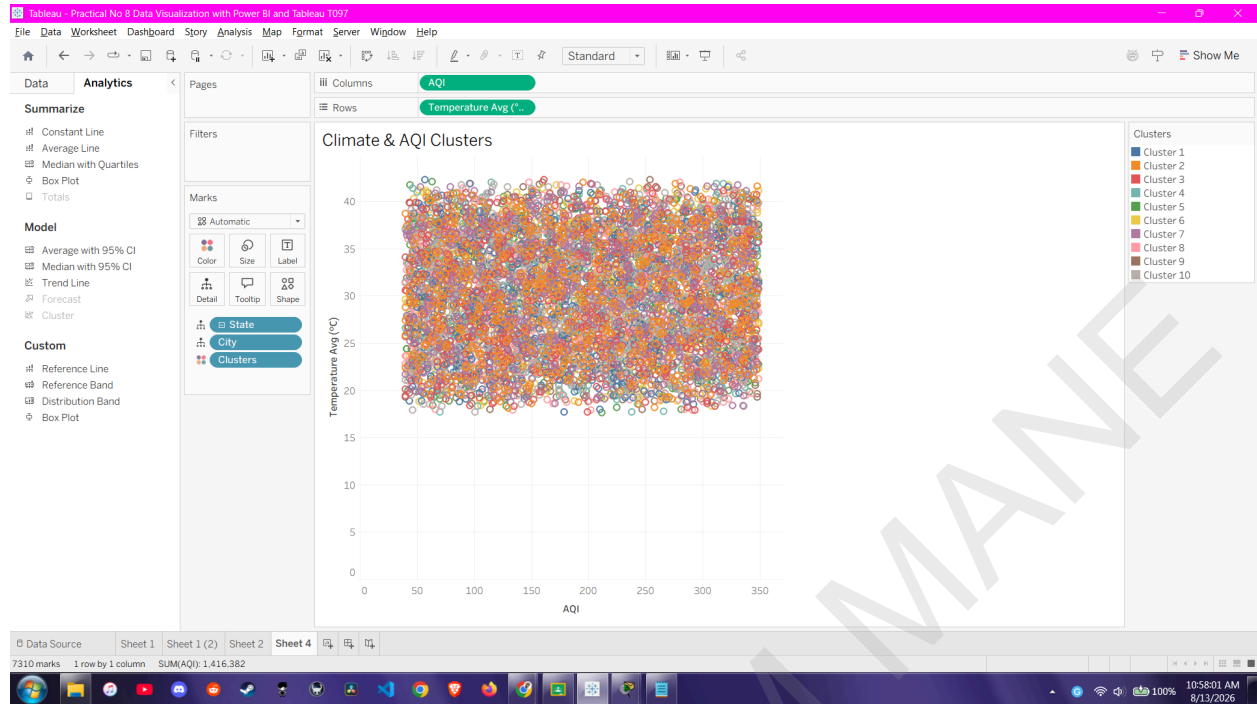
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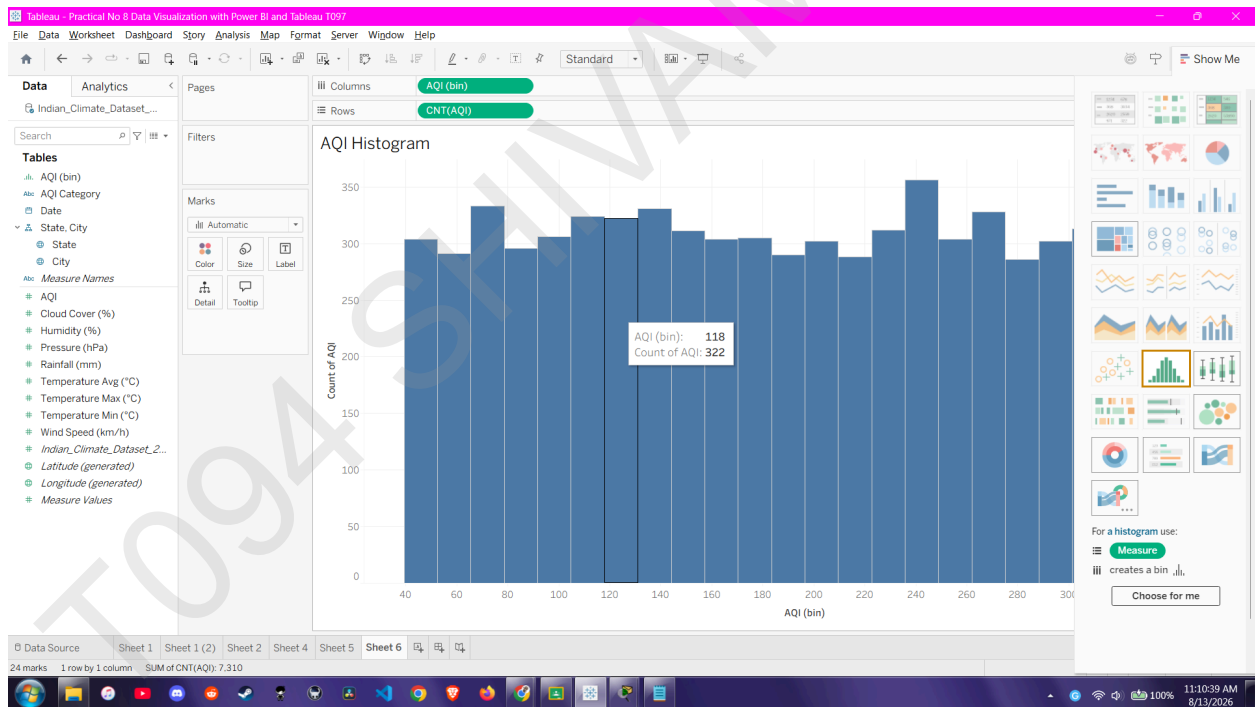
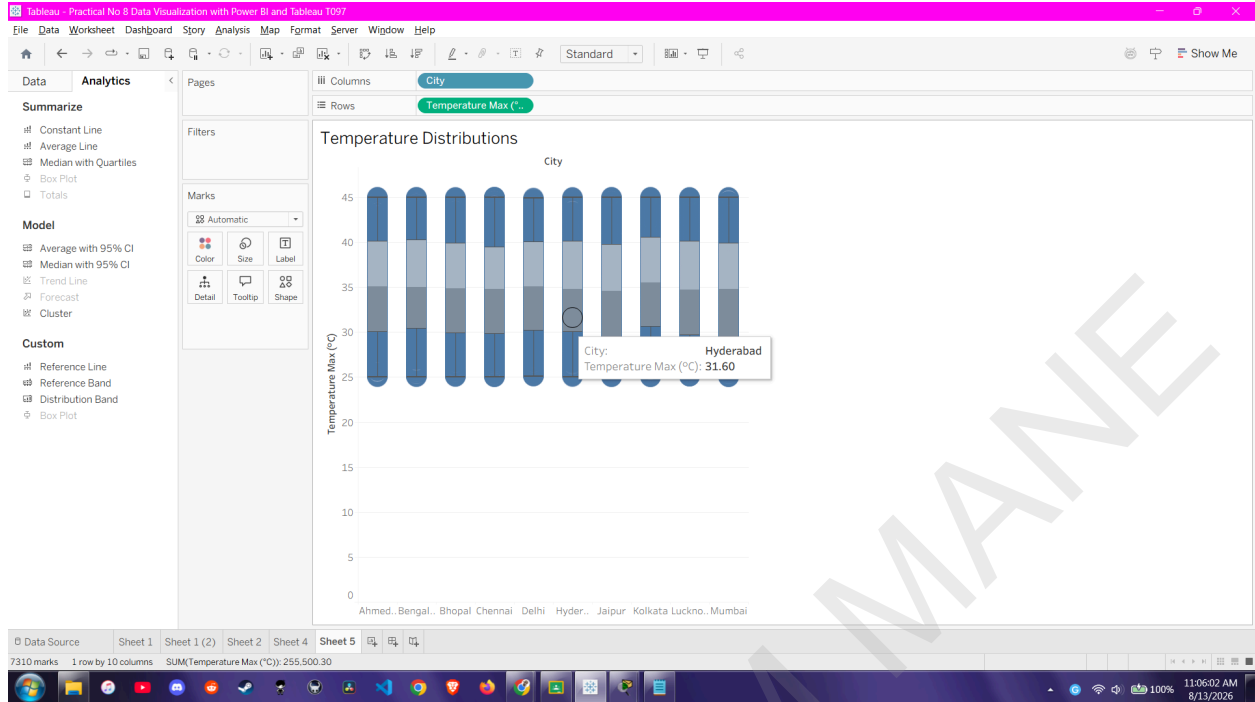
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